ORIENTATION TO COMPUTING

UNIT 4: COMPUTER NETWORKS, TOPOLOGIES & REFERENCE MODELS (OSI & TCP/IP)
Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Data Communications and Networking with TCP/IP Protocol Suite by Behrouz A. Forouzan

>> 4.1 Foundations of Data Communications and Transmission

Data communication is the exchange of data between two devices via some form of transmission medium:
* Five Core Components: Message (data/information), Sender (transmitter), Receiver (sink), Transmission Medium (physical path), Protocol (governing set of rules).
* Data Flow Directionality:
- Simplex: Unidirectional transmission (e.g., traditional broadcast TV, keyboard to CPU).
- Half-Duplex: Bidirectional transmission, but only one direction at a time (e.g., Walkie-talkie).
- Full-Duplex: Simultaneous bidirectional transmission (e.g., modern telephone system, switched Ethernet).
* Network Geographical Scales: Personal Area Network (PAN - Bluetooth ~10 m), Local Area Network (LAN - office building ~1 km), Metropolitan Area Network (MAN - smart city ~50 km), Wide Area Network (WAN - continent/globe).

>> 4.2 Physical Network Topologies

* 1. Mesh Topology: Every node connected to every other node via dedicated point-to-point links. Links required for N nodes: L = N(N - 1) / 2; I/O ports per node = N - 1. Advantages: High fault tolerance, robust privacy. Drawback: Massive cabling cost.
* 2. Star Topology: Each device connected to a central hub/switch via point-to-point link. Highly popular (standard Ethernet LAN). Single cable fault does not affect network; central switch failure halts network.
* 3. Bus Topology: Single shared backbone cable (multipoint) terminated with resistors. Inexpensive; difficult fault isolation; collision prone.
* 4. Ring Topology: Nodes connected in closed loop; tokens circulate unidirectionally. Deterministic delay; single break disrupts entire ring unless dual-ring (FDDI) is used.
* 5. Hybrid Topology: Combination of two or more topologies (e.g., Star-Bus or Tree

topology).

>> 4.3 Transmission Media: Guided vs Unguided

* Guided Media (Conducted physical waves):
- Twisted Pair Cable: Pairs of copper wires twisted to cancel electromagnetic interference (EMI). Unshielded Twisted Pair (UTP Cat6: up to 10 Gbps at 55 m) and Shielded Twisted Pair (STP). RJ-45 connector.
- Coaxial Cable: Central copper core, insulating dielectric, woven copper braid, outer plastic jacket. BNC connector.
- Fiber Optic Cable: Core and cladding transmitting light pulses via Total Internal Reflection (TIR). Immune to EMI; immense bandwidth. Single-mode (long distance laser) vs Multimode (short distance LED).
* Unguided Media (Wireless electromagnetic broadcast): Radio waves (omnidirectional, penetrate walls), Microwaves (line-of-sight parabolic antennas), Infrared (short-range, blocked by walls).

>> 4.4 The 7-Layer OSI Reference Model (ISO 7498)

Theoretical standard model partitioning network communications into 7 hierarchical abstraction layers:
* Layer 7 - Application: User interface, network services (HTTP, DNS, SMTP, FTP).
* Layer 6 - Presentation: Syntax translation, character encoding, data compression, encryption/decryption (SSL/TLS).
* Layer 5 - Session: Dialog establishment, checkpointing, token management, session synchronization.
* Layer 4 - Transport: End-to-end reliable process-to-process delivery, segmentation, flow control, port addressing (TCP, UDP). Data unit: Segment / Datagram.
* Layer 3 - Network: Host-to-host delivery, logical addressing (IP), path determination, routing. Data unit: Packet.
* Layer 2 - Data Link: Hop-to-hop node delivery, physical MAC addressing, framing, error detection (CRC), flow control, channel access (CSMA/CD). Data unit: Frame.
* Layer 1 - Physical: Transmission of raw unstructured bit streams over physical medium, voltage levels, signaling rates, cable connectors. Data unit: Bit.

>> 4.5 The 4-Layer TCP/IP Protocol Architecture

* Practical implementation model powering the modern global Internet:
- Application Layer (OSI Layers 5, 6, 7 consolidated): HTTP, HTTPS, DNS, DHCP, SSH.

- Transport Layer (OSI Layer 4): TCP (connection-oriented) and UDP (connectionless).
- Internet Layer (OSI Layer 3): IP (IPv4, IPv6), ICMP, ARP, routing protocols (OSPF, BGP).
- Network Interface Layer / Link Layer (OSI Layers 1, 2 consolidated): Ethernet, Wi-Fi (802.11).

[PROBLEM 1: Mesh vs Star Topology Enterprise Scalability Calculation]

Problem Statement: An enterprise connects 16 servers in a mission-critical datacenter. Calculate and compare: (1) Total dedicated communication links required for a fully connected Mesh topology vs a Star topology, and (2) Total hardware I/O ports required across all servers.

Mathematical Derivation:

* 1. Mesh Topology Calculations (N = 16):
- Dedicated physical links = N x (N - 1) / 2 = 16 x 15 / 2 = 120 duplex cables.
- I/O ports per server = N - 1 = 15 ports per server.
- Total network I/O ports = 16 servers x 15 ports = 240 ports.
* 2. Star Topology Calculations (N = 16 with central switch):
- Dedicated physical links = N = 16 cables.
- I/O ports per server = 1 port per server (Total = 16 server ports + 16 switch ports = 32 ports).
* 3. Architectural Conclusion: Mesh topology requires 7.5x more physical cabling (120 vs 16) and 7.5x more server ports (240 vs 32), illustrating why Star topology dominates standard LAN architectures.

[PROBLEM 2: Data Link Layer Cyclic Redundancy Check (CRC) Derivation]

Problem Statement: A data link layer sender transmits a 10-bit data frame D = 1010010011. The generator divisor polynomial is G = 10011 (degree r = 4). Calculate the 4-bit CRC remainder (checksum) appended to the transmitted frame, and demonstrate receiver-side verification.

Step-by-Step Modulo-2 Arithmetic Execution:

* 1. Append r = 4 zeros to data frame D: Dividend = 1010010011 0000 (14 bits).
* 2. Modulo-2 Binary Division (XOR operations without borrow):
- Divide 10100100110000 by 10011:
- Step 1: 10100 XOR 10011 = 00111 -> Bring down 1 -> 01111.
- Step 2: Next bit 0 -> bring down 0 -> 11110 XOR 10011 = 01101.
- Step 3: Bring down 0 -> 11010 XOR 10011 = 01001.

- Step 4: Bring down 1 -> 10011 XOR 10011 = 00000.
- Step 5: Bring down remaining bits -> Final remainder = 0101_2.
* 3. Transmitted Codeword: Append remainder (0101) to data: 1010010011 0101.
* 4. Receiver Verification: Receiver divides received codeword 10100100110101 by generator 10011. The remainder is strictly 0000, confirming zero bit corruption during transmission.

[PROBLEM 3: End-to-End OSI Protocol Encapsulation Walkthrough]

Problem Statement: Trace the step-by-step encapsulation process when a user clicks a link in a web browser to download a webpage from a remote web server.
Protocol Flow & Header Attachment Trace:
* 1. Application Layer: Browser generates HTTP GET request. Payload = "GET /index.html HTTP/1.1".
* 2. Transport Layer: Encapsulates HTTP payload inside a TCP Segment by adding TCP Header (Source Port = 54321, Destination Port = 80/443, Sequence Number = 1001, Window Size, Checksum).
* 3. Network Layer: Encapsulates TCP segment inside an IP Packet by adding IP Header (Source IP = 192.168.1.50, Destination IP = 93.184.216.34, TTL = 64, Protocol = 6 for TCP).
* 4. Data Link Layer: Encapsulates IP packet inside an Ethernet Frame by adding Ethernet Header (Source MAC = 3C:52:82:11:22:33, Gateway Router MAC = 00:1A:2B:3C:4D:5E, EtherType = 0x0800) and Ethernet Trailer (4-byte CRC/FCS).
* 5. Physical Layer: Transduces Ethernet frame bits into Manchester electrical pulses / optical photons onto the physical transmission line.

[PROBLEM 4: Bandwidth-Delay Product & Transmission Delay Analysis]

Problem Statement: A 100 Mbps fiber optic communication link spans a transcontinental distance of 4000 km. Signal propagation speed in fiber is 2 x 10^8 m/s. Calculate: (1) Propagation Delay (t_prop), (2) Transmission Delay (t_trans) for a 2.5 MB file, and (3) Bandwidth-Delay Product (BDP) in bits and bytes.
Mathematical Derivation:
* 1. Propagation Delay: t_prop = Distance / Speed = (4000 x 10^3 m) / (2 x 10^8 m/s) = 0.02 seconds = 20 milliseconds (ms).
* 2. Transmission Delay: File size = 2.5 MB = 2.5 x 10^6 x 8 bits = 20,000,000 bits. Bandwidth = 100 Mbps = 100 x 10^6 bps. t_trans = Size / Bandwidth = 20 x 10^6 / 100 x 10^6 = 0.20 seconds = 200 ms.

* 3. Bandwidth-Delay Product (BDP):
- BDP represents the total volume of data bits that can be "in flight" on the transmission wire simultaneously.
- BDP = Bandwidth x Propagation Delay = (100 x 10^6 bps) x 0.02 s = 2,000,000 bits = 250,000 Bytes = 250 KB.
* 4. Engineering Impact: To achieve 100% link utilization, the TCP sliding window size must be configured to at least 250 KB.